

In the Name of God, the Creator of All Crafts

Management Information Systems (MIS) Project

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1 Phase 1 Objectives: Process Simulation & Visual Modeling

Following the initial analytical formulation established in Phase 0, Phase 1 requires the formal graphical representation and technical execution of business workflows using **Visual Paradigm** compliant with the standard **Business Process Model and Notation (BPMN 2.0)** specification.

Core Deliverables & Modeling Scope

1. **Visual Paradigm Project Creation:** Construct complete, syntactically verified BPMN 2.0 diagrams representing the system workflows within the Visual Paradigm environment.
2. **Organizational Partitioning:** Strictly segment all participant roles, departments, and software systems into distinct **Pools** and **Swimlanes**.
3. **Entity Conversion:** Translate all activities, decision nodes, and triggering events documented in Phase 0 into graphical BPMN elements.
4. **Rigorous Metadata Documentation:** Document each task, boundary event, and decision expression with descriptive annotations, data definitions, and condition expressions.

2 Mandatory Technical Constraints & Modeling Rules

To satisfy rigorous enterprise modeling standards, the submitted models must adhere to the following constraints:

- **Specialized Task Typology:** A task represents an atomic unit of work that cannot be subdivided further. General, unspecified tasks must be minimized. **Each modeled process must incorporate at least two (2) specialized task types** (e.g., User Task, Service Task, Script Task, Business Rule Task).
- **Modular Sub-Processes:** To prevent visual clutter and maintain hierarchical abstraction, **each process model must feature at least one (1) Sub-Process** (Embedded Sub-Process, Call Activity, or Transactional Sub-Process).
- **Fault Handling & Boundary Interruption:** Students are expected to implement **Intermediate Boundary Events** (such as Error or Timer events attached to activity perimeters) to model system exceptions, timeouts, and compensation flows.
- **Data & Control Traceability:** Utilize Data Objects, Data Stores, and Text Annotations where informational documents or system databases interface with operational steps.

3 BPMN 2.0 Technical Notation Specification

The following tables define the standard element taxonomy, operational characteristics, and expected utilization in the copper extraction process models.

3.1 1. Task Classifications

Element	BPMN 2.0 Type	Functional Description & Usage
Task	Abstract Task	Default atomic action that cannot be broken down further. Used when task specialization is undefined.
User Task	Human-System Interaction	Performed by a human operator interfacing with an application/BPMS (e.g., operator entering scale data into dispatch terminal).
Service Task	Automated Service	Executed automatically by the system. Interacts with external software services, such as calling RESTful Web APIs.
Receive Task	Message Ingestion	Waits for an external message to arrive from another process or participant before continuing.
Send Task	Message Emission	Dispatches a formal message to an external system or process pool. Acts as a sending communication event.
Script Task	Scripted Routine	Executed directly by the business process engine executing an automated internal code script.
Manual Task	Physical Activity	Purely physical or mechanical operation executed without any BPMS or automated software involvement.
Business Rule	Decision Logic	Introduced in BPMN 2.0 to delegate input parameters to a business rule engine for dynamic evaluation.

3.2 2. Sub-Process Classifications

Element	BPMN 2.0 Type	Functional Description & Usage
Embedded Sub-Process	Compound Activity	A self-contained process embedded wholly inside a parent process, encapsulating internal activities, gateways, and local events.
Call Activity	Reusable Process	References a predefined, globally accessible external process or task definition, promoting process reusability.
Transactional Sub-Process	ACID Transaction	A sub-process controlled by transaction protocols requiring all steps to succeed or triggering coordinated compensation/cancellation.

3.3 3. Gateway Classifications (Branching & Merging Logic)

Element	BPMN 2.0 Type	Functional Description & Usage
Exclusive (XOR)	Conditional Routing	Divergence: routes sequence flow to exactly one mutually exclusive branch. Convergence: merges alternative incoming paths.
Parallel (AND)	Concurrent Execution	Divergence: triggers all downstream concurrent branches without conditions. Convergence: synchronizes and waits for all inbound paths.
Inclusive (OR)	Multi-path Routing	Divergence: triggers one or more conditional paths based on logical evaluations. Convergence: synchronizes all activated branches.
Event-Based	Event-Driven Branching	Branching relies on race conditions between occurrences of subsequent events (e.g., timeout vs. receipt of message).
Complex	Advanced Synchronizer	Controls complex decision patterns and synchronization logic that cannot be expressed by basic AND/XOR combinations.

3.4 4. Event Classifications (Start, Intermediate, Boundary, End)

Element	Event Trigger	Functional Description & Usage
None Start	Generic Initiation	Basic start node indicating process initiation without an explicit trigger.
Message Start	Message Trigger	Initiates a workflow upon the arrival of an external electronic transmission.
Timer Start	Temporal Trigger	Initiates process on a cyclical schedule or specific date/time.
Signal Start	Broadcast Trigger	Initiates process when a global broadcast signal is detected.
Conditional Start	Rule Trigger	Initiates workflow as soon as a predefined business condition evaluates to true.
Inter. Message	Message Handshake	Pauses workflow to wait for an incoming message or emits an outgoing message.
Inter. Timer	Delay / Pause	Enforces a scheduled pause until a specified time interval has elapsed.
Inter. Link	Off-page Connector	Pairs two process segments to avoid lengthy and complex sequence flows.
Inter. Signal	Signal Transceiver	Broadcasts a signal to all listening scopes or waits to catch an active broadcast.
Boundary Error	Exception Catching	Catches errors thrown inside an activity, aborting it to run an exception routine.
Boundary Timer	Timeout Control	Acts as an SLA timeout; triggers alternative action if task exceeds duration.
Boundary Cancel	Transaction Abort	Attached strictly to Transactional Sub-Processes to trigger immediate cancellation.
Boundary Comp.	Compensation Flow	Triggers compensating actions to unwind previously executed transactions.
None End	Final Termination	Normal termination of a specific control path.
Message End	Terminal Message	Concludes process execution while dispatching an outbound transmission.
Error End	Fault Generation	Terminates active execution and raises a named error for catching boundaries.
Cancel End	Transaction Cancel	Signals formal cancellation within a transaction scope.
Terminate End	Immediate Annihilation	Immediately terminates all active tokens and paths across the entire process pool.

3.5 5. Connectors, Artifacts, and Activity Markers

Category	Notation ment	Ele-	Functional Description & Usage
Connectors	Sequence Flow		Solid directed arrows indicating standard execution progression within a pool.
	Message Flow		Dashed arrows modeling inter-pool communications between disparate participants.
	Association		Dotted lines linking text annotations or data objects to process elements.
Artifacts	Data Object		Models informational payloads, physical documents, or database records.
	Text Annotation		Provides human-readable contextual documentation attached to elements.
	Group		Visual dashed bounding box grouping elements for analytical clarity.
Activity Markers	Loop Marker		Indicates that an activity repeats sequentially until a condition evaluates to false.
	Multi-Instance (Par.)		Instantiates parallel concurrent instances of a task across a dataset collection.
	Multi-Instance (Seq.)		Instantiates sequential consecutive instances of a task across a collection.
	Ad-Hoc Marker		Marks a sub-process containing tasks that execute in arbitrary or optional sequence.
	Compensation Marker		Designates an activity dedicated solely to reversing the effects of an action.